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Automated Confession Booth System

Final Requirement in the course
CPE319 – Elect 1 – Embedded System 1

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I. Introduction

1. Background and Problem

The traditional confessional booth has served as a vital medium for religious individuals to seek forgiveness and guidance. However, these systems often rely on manual operation and lack technological integration, which can hinder the user experience. Privacy concerns, inefficient signaling of booth availability, and manual acknowledgment processes pose challenges in modern settings. There is a need to enhance the functionality and accessibility of these booths without compromising their sacred purpose.

2. Rationale

Automation has the potential to modernize religious practices by addressing key issues of efficiency and privacy. By introducing an automated system, this project seeks to provide a seamless confessional experience for users and clergy alike. Incorporating motion detection, status indicators, and minimal manual inputs ensures a streamlined process that respects tradition while embracing innovation.

3. Objectives

The primary goals of the Automated Confessional Booth System are:

- To automate the booth's availability detection using sensors and LED indicators.
- To reduce wiring complexity and reliance on manual inputs.
- To enhance privacy and efficiency during confessional sessions.
- To provide a scalable solution for modern churches seeking technological upgrades.

4. Expected Results and Benefits

The project aims to deliver a fully functional prototype capable of automating confessional processes. Benefits include improved user privacy, reduced workload for



clergy, and a more intuitive experience for users. The modular design ensures adaptability to varying church requirements.

5. Literature Review

Introduction

The integration of technology into religious practices is rapidly evolving, offering new opportunities for accessibility, privacy, and automation. This literature review synthesizes findings from studies on artificial intelligence (AI) in religious practices and IoT-based automation systems, providing insights into the design and development of an automated confession booth system. Relevant works include Rey Ty's exploration of AI's impact on religious practices and the IoT-based home automation project by Manikandan et al., both of which contribute to the conceptual and technical framework of the system.

AI and Religious Practices

Impact of AI in Religion (Ty, 2023). Rey Ty's research highlights the transformative potential of AI in religious settings, such as virtual assistants for counseling and digital confessional systems. Key points include:

1. AI Integration:

-AI-based tools enable private, on-demand access to spiritual guidance and confessions. Examples of AI-driven chatbots, such as AI Jesus and virtual confession tools, provide precedent for automated systems.

2. Ethical Considerations:

-Issues of privacy and data security are critical when handling sensitive religious data. Ethical guidelines are necessary to maintain doctrinal fidelity and ensure the sanctity of spiritual experiences.



3. Positive Impacts:

- Increased accessibility for users unable to attend physical locations.
- Preservation of traditional practices through innovative means.

4. Negative Impacts:

- Potential for generic, impersonal responses lacking human empathy.
- Over-reliance on automation, leading to reduced community engagement.

This research underscores the importance of maintaining ethical practices while leveraging AI to enhance privacy and efficiency in religious observances.

IoT-Based Automation Technologies

IoT-Based Home Automation (Manikandan et al., 2020). Manikandan and colleagues developed a home automation system using PIR motion sensors and Node MCU (ESP8266). The system's architecture provides valuable technical insights applicable to an automated confession booth:

1. PIR Motion Sensors:

- Detects motion by sensing infrared radiation.
- Ideal for detecting when a penitent enters the booth, triggering system activation.

2. . Node MCU Integration:

- Compact and Wi-Fi-enabled microcontroller for managing inputs and server communication.
- Allows real-time notifications to priests, enabling remote monitoring.



3. **Arduino Integration:**

- While Node MCU is highlighted in the study, Arduino Uno serves as a suitable alternative due to its simplicity and accessibility.
- Enables integration with PIR sensors and relay mechanisms for seamless automation.

4. **Relay Mechanisms:**

- Controls devices like lights and fans based on sensor inputs.
- Could manage booth lighting or activate privacy features automatically.

5. **Mobile Application Connectivity:**

- Integration with apps like Blynk demonstrates how remote control and monitoring can be achieved.
- Enables priests to receive alerts or view booth activity via mobile interfaces.

6. **Software Development:**

- Arduino IDE provides a user-friendly environment for programming the Node MCU.
 - Logical flow ensures seamless operation of detection and response systems, essential for automating confessional processes.
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Applications to the Automated Confession Booth System

Combining insights from these studies, the automated confession booth system can:

- Utilize **PIR motion sensors** to detect booth entry and trigger automated processes.
- Employ **Node MCU or ESP32** for network telecommunications.
- Employ **Arduino Uno** for an alternative yet reliable communication and control.
- Implement **ethical AI** features to safeguard sensitive data and enhance user privacy.



-
- Develop a mobile application or a simple notification system to alert priests of booth activity.

Conclusion

The integration of AI and IoT technologies into religious practices represents a significant advancement in accessibility and functionality. Ty's exploration of AI in religion and Manikandan et al.'s technical contributions form the foundation for developing an automated confession booth system. While the potential benefits are substantial, careful attention to ethical considerations and privacy safeguards will ensure the system aligns with the spiritual integrity and requirements of confessional practices.



II. Materials and Methods

1. List of Hardware Materials

Component	Purpose	Quantity
Arduino Uno	Microcontroller to control the system	1
LCD w/ I2C	To display system messages and statuses	1
PIR Sensor	To detect motion within the booth	1
LEDs	To indicate booth statuses (ready, occupied, etc.)	5
Push Buttons	To start and end confession sessions and acknowledge them	4
Resistors 1K	For the stable operation of buttons	4
Resistors 330K	For stable operation of LEDs	5
Buzzer	To provide audio feedback during interactions	2
Dupont Wires	For electrical connections	Multiple
Breadboard	For organizing connections during prototyping	2
Power Supply	To power the Arduino system	1

Table 1 - Table of Hardware Requirements

2. List of Software Requirements

Software Tool	Purpose
Arduino IDE	To program the Arduino and upload the sketch
Liquid Crystal_I2C Library	For controlling the I2C-based LCD module
Serial Monitor	To debug and monitor system outputs during testing

Table 2 - Table of Software Requirements



3. Project Architecture or Diagrams

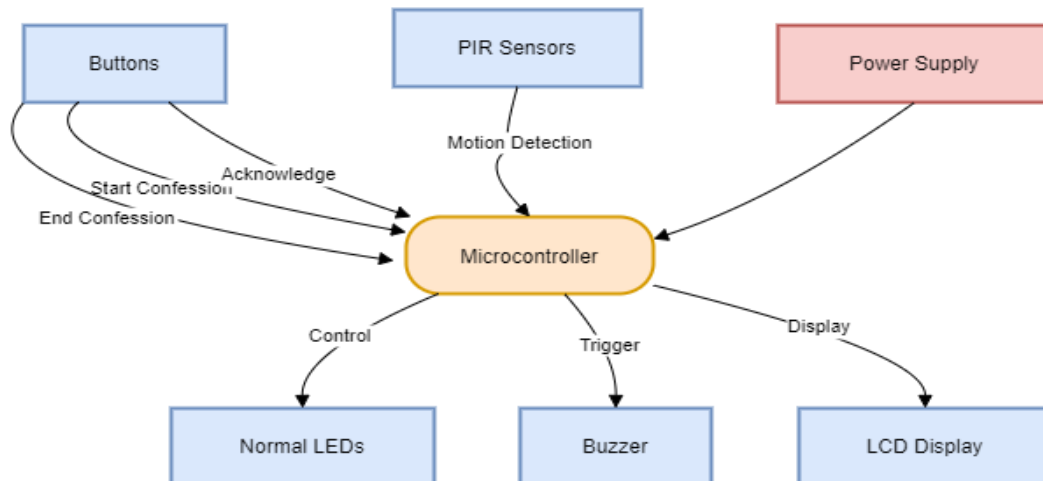


Figure 1 - Block Diagram of Automated Confession Booth System

Description of Components

I. Buttons:

One for starting the confession, one for the priest's acknowledgment, and one for ending the confession.

II. PIR Sensors:

Monitors movement and provides additional detection capabilities.

III. Microcontroller:

Processes inputs and controls outputs based on the logic programmed.

IV. Normal LEDs:

Indicate the status of the booth (e.g., available or busy).

V. Buzzer:

Provides auditory signals for notifications.

VI. LCD Display:

Shows instructions or messages to users.

VII. Power Supply:

Supplies power to the entire system



4. Assembling Procedures and Flowchart

Gather Materials: Collect all components: microcontroller (Arduino), PIR sensors, ultrasonic sensors, buttons, normal LEDs, a buzzer, and an LCD.

Prepare the Breadboard: Arrange the components on the breadboard or PCB for easy access and organization.

Connect the Sensors: Connect the PIR and ultrasonic sensors to the appropriate pins on the microcontroller. Ensure power and ground connections are properly established.

Install Buttons: Connect the confession start button and the acknowledgment button to designated input pins on the microcontroller.

Wire the LEDs: Connect normal LEDs to the output pins of the microcontroller. Include current-limiting resistors as needed.

Add the Buzzer: Connect the buzzer to the microcontroller, ensuring it can be activated via a specific output pin.

Set Up the LCD Display: Wire the LCD to the microcontroller using the appropriate pins for data and control.

Power Connections: Ensure all components are connected to a suitable power source.

Upload Code: Write and upload the program code to the microcontroller to define how it interacts with all components.

Testing: Power on the system and test each component to ensure proper functionality.

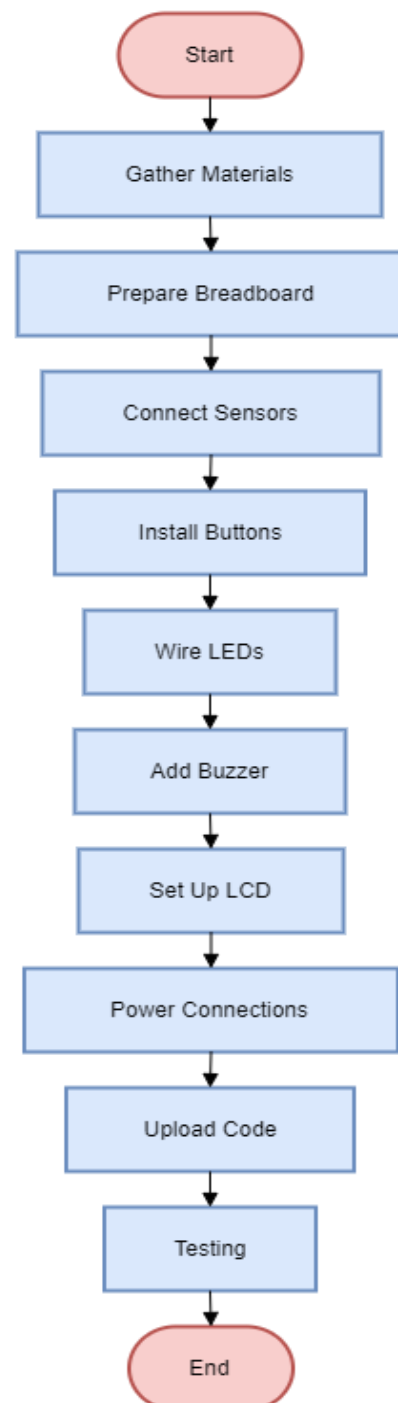


Figure 2 - Assembly's Flowchart



5. Program Flowchart

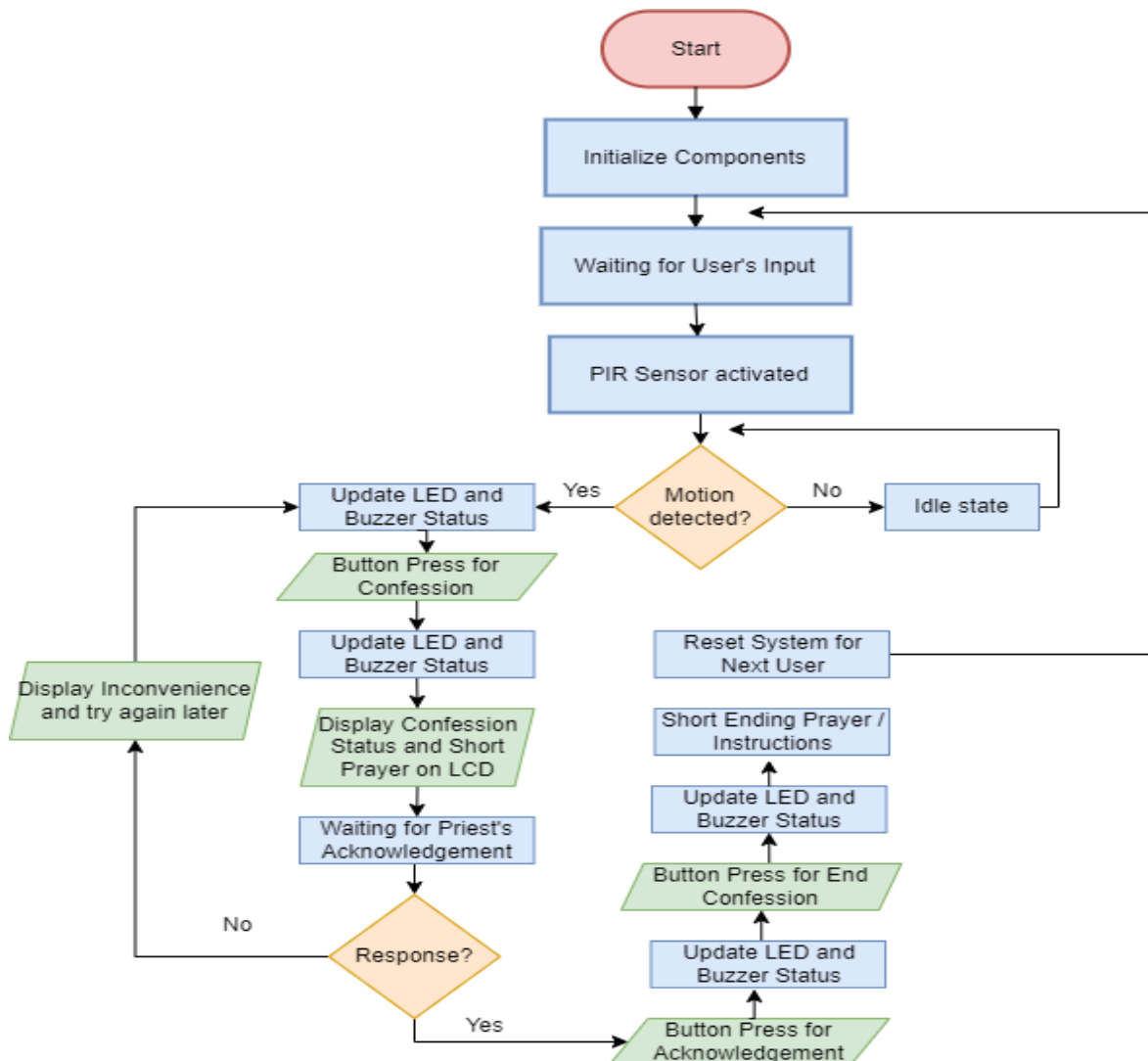


Figure 3 - Program's Flowchart

Start: The program begins execution.

Initialize Components: All hardware components (sensors, buttons, LEDs, LCD) are initialized. This part is crucial to make the PIR Sensor work.

Waiting for User Input: The program is now ready for the button to be pressed.



PIR Sensor Activated: PIR Sensor is now ready to be used

Motion Detected: If motion is detected, it proceeds to update LED Status and Buzzer Status. If not, it will go to the idle state, waiting for a motion to be detected.

Button Press for Confession: This is where, after motion is detected and the LED and Buzzer Status are updated, they can start the confession through the button.

Display Short Prayer through LCD: A short prayer before the start of confession.

Wait for Acknowledgment from Priest: The system waits for the priest's acknowledgment.

Response: If there is no response, the system will try to message the user about the inconvenience and try again later. If there is a response, the priest will press the acknowledgement button.

End Confession (On the Penitent's side): If done, they can press this button to end the confession.

Play Buzzer or Prayer Instructions: Plays a sound or gives instructions for the next steps.

Reset System for Next User: Prepares the system for the next user.

Loop Back: The program returns to checking for user input again.

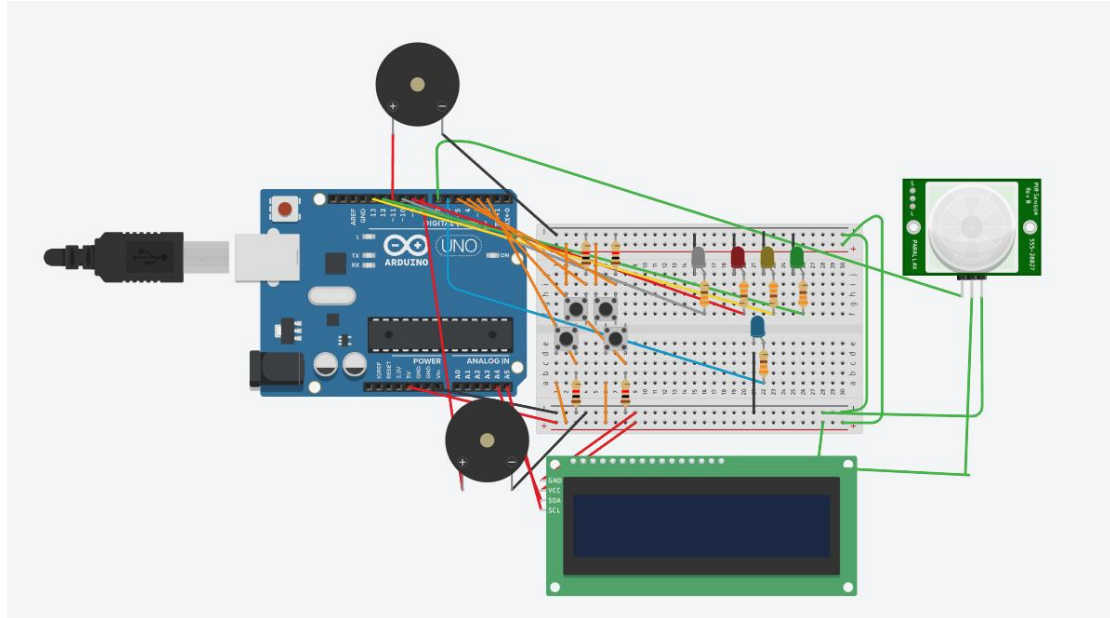


Figure 4 – Circuit Diagram

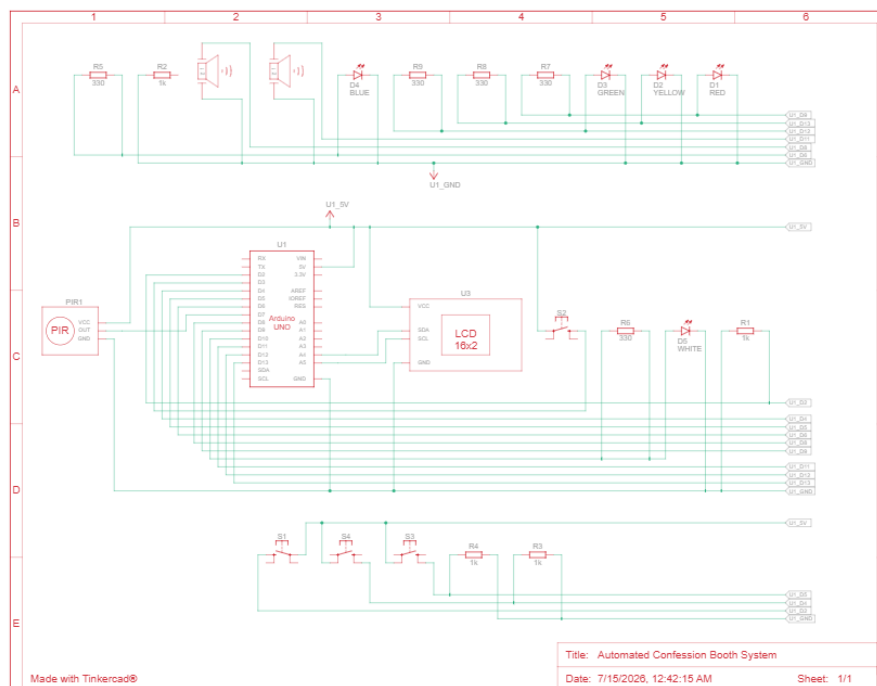


Figure 5 –Schematic Diagram



6. Expected Screen Output

For this Screen Output. I will highlight what users expect to see on the LCD screen during different stages of interaction on the Automated Confession Booth System

1) Startup Screen:

Message: "Automated Confession Booth"

Duration: Displayed for 2 seconds during startup.

2) Calibration Screen:

Message: "Calibrating Sensor..."

Duration: Shown while the calibration process is happening.

3) Ready Screen:

Message: "Booth is Ready"

Duration: Displayed after calibration is complete, indicating that the booth is ready for use.

4) Confession Start Screen:

Message (General Confession): "General Confession"

Message (Specific Confession): "Specific Confession"

Duration: Displayed when the user presses the button to start a confession.

5) Reflection Prompts:

Message (e.g., "Pause, reflect...")

Duration: Each prompt was displayed for a few seconds during the confession process.



6) Confession Ongoing Screen:

Message: "Confession is Ongoing..."

Duration: Displayed after the priest acknowledges the confession.

7) Conclude Confession Screen:

Message: "Confession Done"

Duration: Displayed briefly after the user ends the confession session.

Booth Available Screen:

Message: "Booth Available"

Duration: Displayed after a confession is concluded or if the priest is unavailable.



III. Results and Discussion

1. Project Description

Automated Confession Booth Overview:

The Automated Confession Booth is a system designed to facilitate a private and structured confession experience for users. Utilizing sensors and simple user interfaces, the booth allows penitents to start a confession, while a priest can acknowledge and conclude the session through the use of LEDs and buttons. The project aims to modernize the traditional confession process, making it more efficient and respectful of users' privacy. Key components include:

- **PIR Sensors:** To detect the presence of individuals in the booth through motion.
- **Buttons:** To start and end confession sessions.
- **LED Indicators:** To provide status updates (available, busy, etc.).
- **LCD Display:** To show relevant messages during the confession process.

This project leverages embedded systems technology, combining hardware and software elements to create an interactive and user-friendly experience.

2. 3D of the Prototype

The Automated Confession Booth is designed as a compact, enclosed space that provides privacy for users while maintaining a welcoming environment. The booth features:



- **Dimensions:** Approximately 1.5 meters in height, 1 meter in width, and 1 meter in depth.
- **Material:** Something simple like plastic or cardboard will do.
- **Interior Setup:**
 - **Control Panel:** Equipped with buttons for starting and ending confession sessions and LED indicators for status updates.
 - **LCD Display:** Positioned at eye level to provide guidance and prompts during the confession process.
 - **PIR Sensors:** Installed at the entrance to detect motion and presence within the booth.

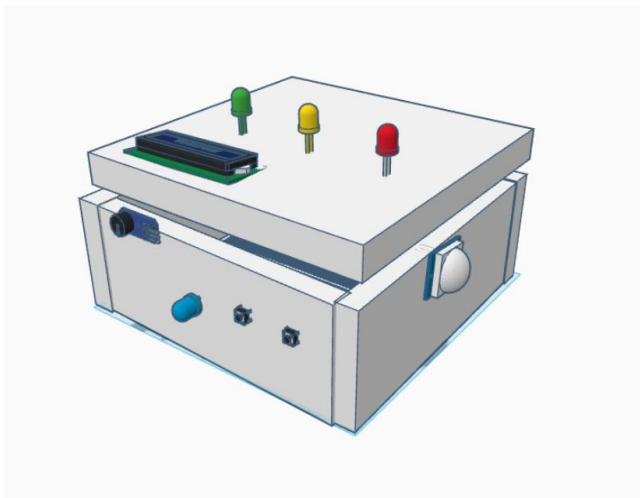


Figure 6.1 - 3D Prototype Front

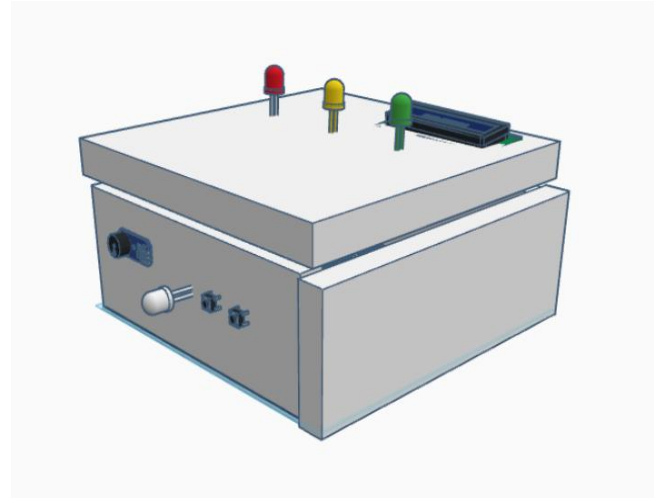


Figure 6.2 - 3D Prototype Back



3. Actual Prototype Images

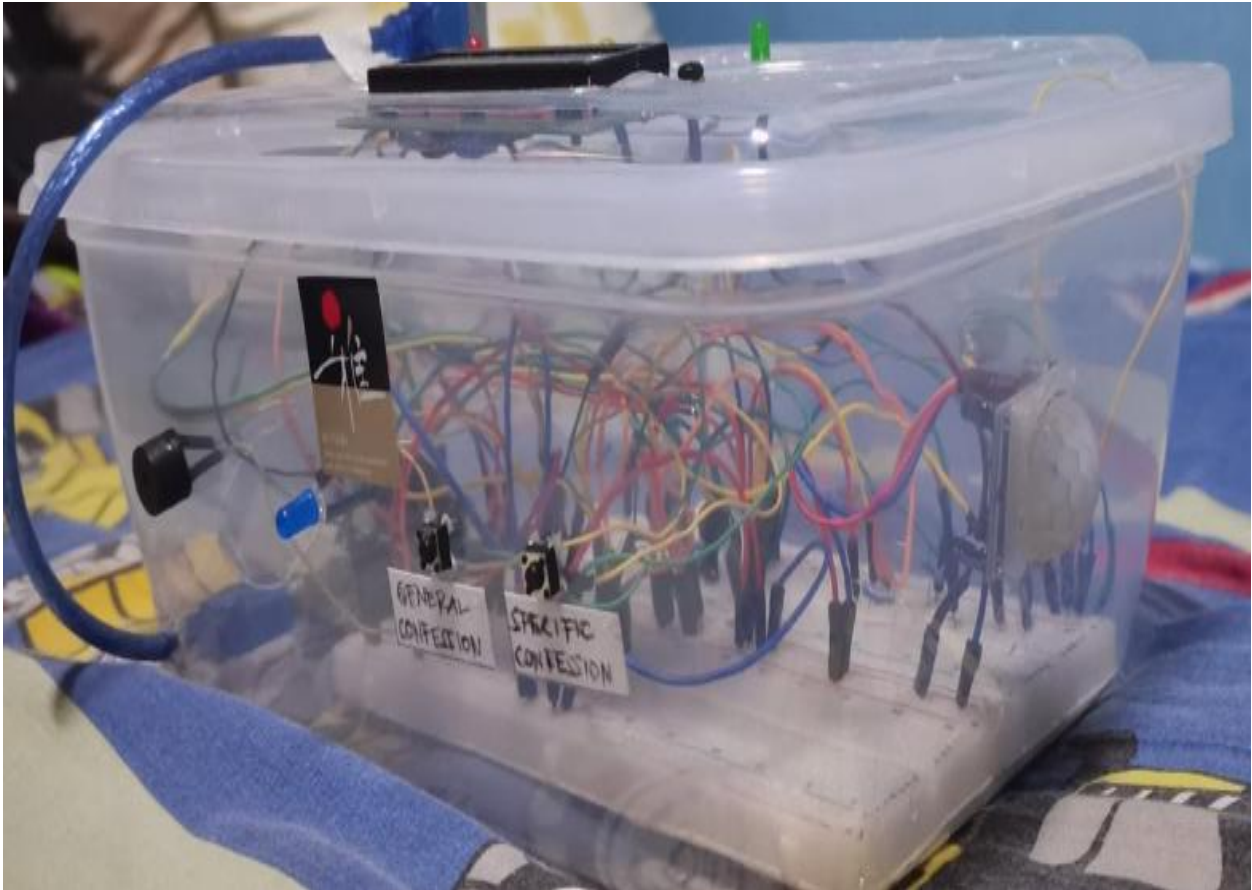


Figure 7.1- Prototype Front

Front Side (Penitent's Perspective):

The front side represents the penitent's perspective. Here, the penitent presses a button to initiate the confession process. A buzzer and blue LED, along with the PIR sensor positioned on the right side of the container, simulate the penitent's entrance into the booth. When the PIR sensor detects motion, it updates the system status by activating the buzzer and blue LED, signaling the start of the session.



Figure 7.2- Prototype Back

The back side represents the priest's perspective. The priest has access to a button used to acknowledge the penitent's confession. The notification system includes a buzzer and a white LED. The buzzer alerts the priest of the confession, while the white LED illuminates to indicate that a reply or action is expected.

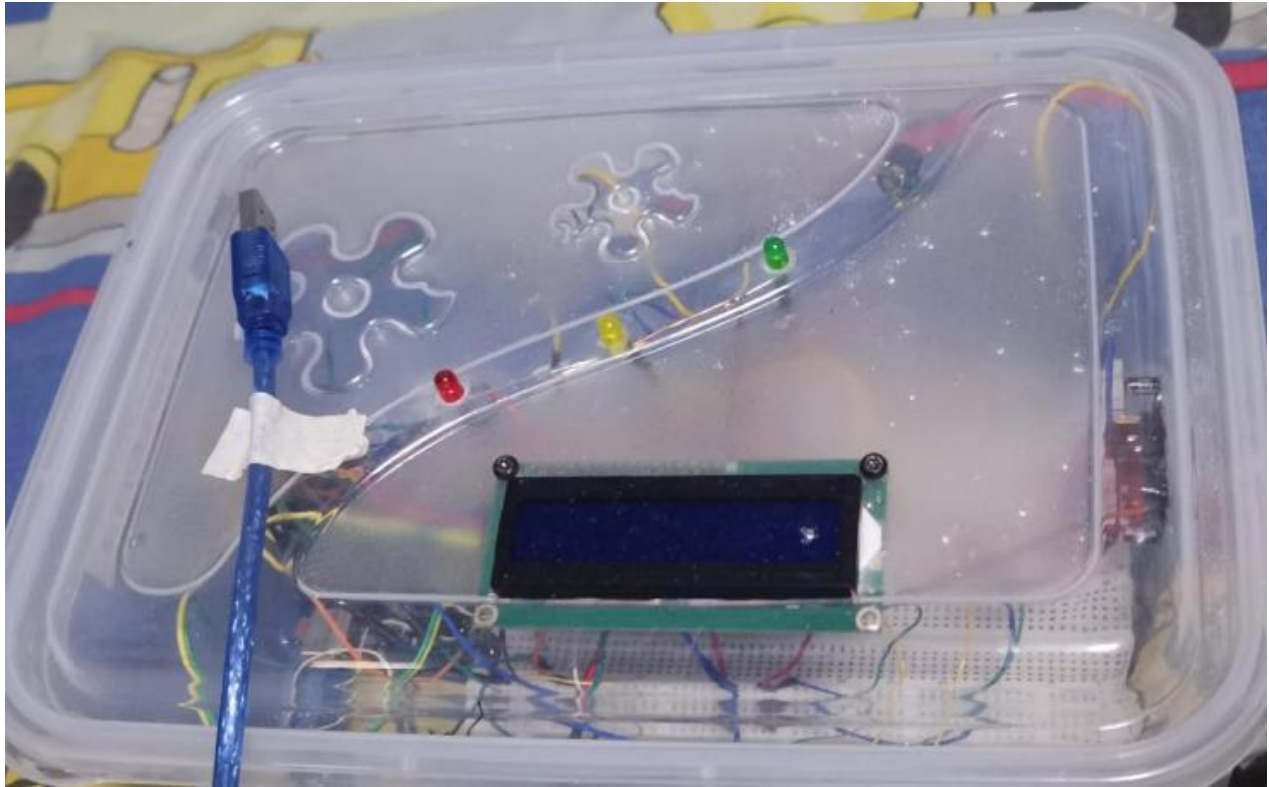


Figure 7.3 - Prototype Top

Top View (LED Status Indicators):

The top view showcases the booth's LED status system:

- **Red LED:** Indicates the booth is occupied, and a confession is ongoing.
- **Yellow LED:** Signals that a session is about to start.
- **Green LED:** Denotes the booth is available for use.



4. Testing Results

Trials	Input	Output	Results
1.	Press General Confession	LEDs Update to yellow, Buzzer sounds once, LCD Display prayer	Correct Output: System Updates perfectly
2.	Press Specific Confession	LEDs Update to yellow, Buzzer sounds once, LCD Display prayer	Correct Output: System Updates perfectly
3.	Motion detected by PIR sensor (penitent)	Buzzer sounds, Blue LED remains ON	Correct Output: Motion triggers correct notification.
4.	Priest presses acknowledgment button	White LED OFF, Buzzer sounds once, Red LED on	Correct Output: Priest notified successfully.
5.	Priest presses button (end session)	Status LED changes to GREEN, Buzzer sounds once	Correct Output: Confession session ends correctly.
6.	No motion detected after a timeout period	PIR System reverts to IDLE state	Correct Output: PIR System transitions to idle properly.
7.	PIR sensor detects false motion (ambient)	Response	Incorrect Output: False motion detected
8.	Simultaneous button press (both sides)	No interruption, system prioritizes flow	Correct Output: System handles simultaneous inputs.
9.	Manually block PIR sensor (obstruction)	No detection, LEDs unchanged	Correct Output: System works only with real motion.
10.	PIR sensor detects fast moving motion.	Sometimes response, sometimes it don't.	Partially Correct: It still detects motion but needs further calibration.

Table 3 - Testing Results of Automated Confession Booth System



5. Errors encountered and Solution done

- ***False Motion Detection***

Issue: The PIR sensor occasionally detected false motions caused by ambient factors, leading to unnecessary notifications.

Solution: Adjustments to the sensor's positioning and sensitivity were explored. Future iterations may include integrating shielding or alternative sensor configurations.

- ***Inconsistent Motion Detection for Fast Movements***

Issue: The PIR sensor had difficulty consistently detecting fast-moving objects.

Solution: Calibration of the sensor's settings is planned to improve detection accuracy.

- ***Excessive Wiring and Component Complexity***

Issue: The initial design involved a significant number of wires, resulting in a cluttered and less efficient setup.

Solution: Simplification is ongoing, with multiplexing techniques and reduced components being explored to streamline the design. Additionally, integrating a NodeMCU ESP module is proposed as a future solution. This would help reduce wiring by enabling wireless communication and minimizing physical connections. However, further study and testing of the NodeMCU ESP are required before implementation.

- ***Prototype Aesthetics***

Issue: The prototype's housing was constructed using a plastic tupperware container, which lacked a polished appearance.



Solution: Future improvements will involve creating a durable, visually appealing case using wood or cardboard.

- ***Enhancing Motion Detection for Both Sides***

Issue: Detection was limited to the penitent's side. Adding detection for the priest's side was suggested.

Solution: This is deferred for now due to hardware limitations but will be considered for future updates.

- ***Simultaneous Button Press Handling***

Issue: Potential conflicts could arise from simultaneous button presses by the penitent and priest.

Solution: The current system prioritizes input flow, ensuring no interruptions during operations.

- ◆ ***Unused Components***

- **SD Card:** Planned for data storage but not integrated due to time constraints and lack of need in the final design.
- **Logic Shifter:** Intended for voltage level adjustments, but ultimately deemed unnecessary for the current circuit design.
- **NodeMCU 8266:** Considered for adding wireless capabilities; however, integration was postponed due to unfamiliarity and time constraints.



IV. Conclusion and Recommendation

Conclusion:

The development of the automated confessional booth successfully demonstrated the integration of embedded systems into real-world applications. The prototype achieved its primary objectives of providing an accessible and semi-automated confession process by utilizing motion sensors, LEDs for status indication, and buttons for user interaction. Key features such as real-time motion detection, system status updates, and acknowledgment notifications effectively simulated the intended functionalities. Despite some minor issues, including occasional false motion detections and challenges in the aesthetic quality of the prototype, the overall system performed as expected.

The project highlights the potential of embedded systems to enhance traditional practices with modern technology. It also underscores the importance of iterative design, testing, and user feedback in refining and improving system performance.

Recommendation:

- I. ***Enhancing Motion Detection:*** Upgrade the PIR sensor or consider using alternative sensors (e.g., dual PIR sensors or ultrasonic sensors) to improve detection reliability and accommodate motion detection from both penitent and priest sides.
- II. ***Streamlining Components:*** Further reduce the number of wires and hardware complexity by integrating more efficient wiring systems or exploring wireless communication modules.
 - ◆ Considering integrating the NodeMCU ESP module in future iterations of the project. This upgrade could significantly streamline the circuit by reducing wiring complexity, offering wireless communication



capabilities, and providing a more compact and efficient design. While this requires additional learning and testing, it could greatly improve the system's functionality and maintainability."

- III. ***Improving Prototype Aesthetics:*** Replace the temporary housing material (e.g., plastic tupperware) with a sturdier and more visually appealing casing .
- IV. ***System Calibration:*** Refine sensor sensitivity and timing settings to minimize false detections and enhance responsiveness.
- V. ***Future Expansions:*** Add features such as audio feedback or a digital screen to display prayers or instructions for better user interaction and accessibility.
- VI. ***Documentation and Scalability:*** Provide detailed user manuals and guidelines for scaling the project to other applications or venues.

These recommendations aim to refine the design further and expand the functionality of the automated confessional booth for broader use and acceptance.



References

- Ty, Rey. (2023). *Impact of AI-Powered Technology on Religious Practices and Ethics: The Road Ahead*. Religion and Social Communication, Vol. 21 No. 2.
- Manikandan, M., Ranjithkumar, R., Rathish Ganesh, S., & Ram Vikash, K. (2020). *IoT Based Home Automation using PIR Motion Sensor and Node MCU*. International Journal of Engineering and Advanced Technology (IJEAT), Vol. 9 Issue 4.

Appendix A - Proposed Budget of the Project

Item	Quantity	Unit Price	Total Price
Arduino UNO R3	1	324 Pesos	324 Pesos
Dupont Wires 10CM	40	30 Pesos	160 Pesos
HC-SR01 PIR	1	52 Pesos	52 Pesos
Node MCU ESP8266	1	288 Pesos	348 Pesos
Micro SD Card Module	1	35 Pesos	95 Pesos
Bi Directional Logic Shifter	1	68 Pesos	168 Pesos
Total			1,147 Pesos

Table 4 - Budget Price from Online Shopping via Shopee



Appendix B - Project Timeline

Task	Start Date	End Date	Duration
Research and Planning	Oct. 10, 2024	Oct. 17,2024	7 days
Component Procurement	Oct. 30, 2024	Nov. 13, 2024	14 days
Prototype Development	Nov. 18, 2024	Dec. 3, 2024	15 days
Testing and Evaluation	Dec. 5, 2024	Dec. 7, 2024	3 days
Final Documentation	Dec. 10, 2024	Dec. 14, 2024	5 days
Total			44 Days

Table 5 - Detail Timeline for Project's Progress

Appendix C - Program Source Code

```
#include <Wire.h>
#include <LiquidCrystal_I2C.h>

// Define pins (same as before)
#define BUTTON1_PIN 2
#define BUTTON2_PIN 3
#define BUTTON3_PIN 4
#define BUTTON_ACKNOWLEDGE_PIN 5

int pirPin = 7;
int ledPin = 6;
int Buzzer = 8;

int priestLedPin = 10;
int priestBuzzerPin = 11;

int statusGreenPin = 12;
int statusYellowPin = 13;
int statusRedPin = 9;

LiquidCrystal_I2C lcd(0x27, 16, 2);

int calibrationTime = 30;
boolean penitentWaiting = false;
unsigned long confessionStartTime = 0;
boolean timerActive = false;
```



```
boolean personInBooth = false;  
unsigned long lastMotionDetectedTime = 0;  
const unsigned long motionTimeout = 10000; // 10 seconds of no motion = booth is vacant
```

```
void setup() {  
  Serial.begin(9600);  
  lcd.begin();  
  lcd.backlight();  
  lcd.print(" Automated");  
  lcd.setCursor(0, 1);  
  lcd.print("Confession Booth");  
  delay(2000);  
  lcd.clear();  
  
  // Turn off the green LED before calibration  
  digitalWrite(statusGreenPin, LOW);  
  
  pinMode(BUTTON1_PIN, INPUT_PULLUP);  
  pinMode(BUTTON2_PIN, INPUT_PULLUP);  
  pinMode(BUTTON3_PIN, INPUT_PULLUP);  
  pinMode(BUTTON_ACKNOWLEDGE_PIN, INPUT_PULLUP);  
  
  pinMode(pirPin, INPUT);  
  pinMode(ledPin, OUTPUT);  
  pinMode(Buzzer, OUTPUT);  
  
  pinMode(priestLedPin, OUTPUT);  
  pinMode(priestBuzzerPin, OUTPUT);  
  
  pinMode(statusGreenPin, OUTPUT);  
  pinMode(statusYellowPin, OUTPUT);  
  pinMode(statusRedPin, OUTPUT);  
  
  lcd.print("Calibrating");  
  lcd.setCursor(0, 1);  
  lcd.print("Sensor.....");  
  Serial.print("Calibrating sensor ");  
  for (int i = 0; i < calibrationTime; i++) {  
    Serial.print(".");  
    delay(1000);  
  }  
  Serial.println(" done");  
  lcd.clear();  
  
  // Turn on the green LED after calibration is done  
  digitalWrite(statusGreenPin, HIGH);  
  
  lcd.print(" Sensor Active");
```



```
lcd.clear();
lcd.print(" Booth is Ready");
Serial.println("SENSOR ACTIVE");
delay(50);
}

void loop() {
  checkConfessionButtons();
  checkAcknowledgeButton();
  checkMotionAndBuzzer();
  handleTimer();
  delay(100);
}

void checkConfessionButtons() {
  if (digitalRead(BUTTON1_PIN) == HIGH) {
    startConfession("General Confession");
  }

  if (digitalRead(BUTTON2_PIN) == HIGH) {
    startConfession("Specific Confession");
  }

  // Button to conclude the confession
  if (digitalRead(BUTTON3_PIN) == HIGH) {
    concludeConfession();
  }
}

void startConfession(String confessionType) {
  notifyPriest(confessionType);
  setBoothStatus(1);
  penitentWaiting = true;
  timerActive = true;
  confessionStartTime = millis();

  lcd.clear();
  if (confessionType == "General Confession") {
    lcd.print("General Confess.");
    lcd.setCursor(0, 1);
    delay(2000); // Wait briefly before showing the next message

    // Display reflection prompts
    lcd.clear();
    lcd.print("Pause, reflect...");
    delay(1000);
    lcd.setCursor(0, 1);
    lcd.print("On your journey...");
    delay(3000); // Display message for 3 seconds
```



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```
lcd.clear();
delay(500);
lcd.print("Think of gifts...");
delay(1000);
lcd.setCursor(0, 1);
lcd.print("And Trials....");
delay(3000);
lcd.clear();
delay(500);
lcd.print("Seek healing and...");
delay(1000);
lcd.setCursor(0, 1);
lcd.print("Open your heart...");
delay(3000);
lcd.clear();
lcd.print("Please wait...");

} else if (confessionType == "Specific Confession") {
    lcd.print("Specific Confession");
    lcd.setCursor(0, 1);
    delay(2000);

    lcd.clear();
    lcd.print("Focus on recent...");
    delay(1000);
    lcd.setCursor(0, 1);
    lcd.print("thoughts, deeds...");
    delay(3000); // Display message for 3 seconds
    lcd.clear();
    delay(500);
    lcd.print("Acknowledge your...");
    delay(1000);
    lcd.setCursor(0, 1);
    lcd.print("Problems, Sins...");
    delay(3000);
    lcd.clear();
    delay(500);
    lcd.print("Seek strength to...");
    delay(1000);
    lcd.setCursor(0, 1);
    lcd.print("Move forward....");
    delay(3000);
    lcd.clear();
    lcd.print("Please wait...");
}
}

void checkAcknowledgeButton() {
    if (penitentWaiting && digitalRead(BUTTON_ACKNOWLEDGE_PIN) == HIGH) {
        setBoothStatus(2);
    }
}
```



```
digitalWrite(priestLedPin, HIGH);
tone(priestBuzzerPin, 2200, 500);
Serial.println("Confession ongoing");
lcd.clear();
lcd.setCursor(0, 0);
lcd.print("Confession is");
lcd.setCursor(0, 1);
lcd.print("Ongoing.....");
delay(2000);
digitalWrite(priestLedPin, LOW);
penitentWaiting = false;
timerActive = false; // Stop the timer
}
}

void concludeConfession() {
  setBoothStatus(0); // Booth is now available
  penitentWaiting = false; // Reset waiting state
  timerActive = false; // Stop the timer
  tone(priestBuzzerPin, 2000, 1000);
  lcd.clear();
  lcd.print("Confession Done");
  Serial.println("Confession concluded. Booth is ready.");
  delay(2000); // Brief display for confirmation
  lcd.clear();
  lcd.print("Booth Available");
}

void checkMotionAndBuzzer() {
  static int lastPirState = LOW;
  int pirState = digitalRead(pirPin);

  if (pirState == HIGH && lastPirState == LOW) {
    digitalWrite(ledPin, HIGH);
    tone(Buzzer, 1500, 1000);
    Serial.println("Motion detected");
    lastPirState = HIGH;
  } else if (pirState == LOW && lastPirState == HIGH) {
    digitalWrite(ledPin, LOW);
    noTone(Buzzer);
    lastPirState = LOW;
  }
}

void handleTimer() {
  if (timerActive && (millis() - confessionStartTime >= 30000)) {
    // 30 seconds have passed, and no acknowledgment from the priest
    setBoothStatus(0); // Reset booth to available
    penitentWaiting = false;
  }
}
```



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```
timerActive = false; // Stop the timer
lcd.clear();
lcd.print("Sorry, Priest is");
lcd.setCursor(0,1);
lcd.print("Unavailable now..");
delay(2000);
lcd.clear();
lcd.print("Try again later..");
delay(3000);
Serial.println("Priest unavailable");
delay(2000); // Display message briefly
lcd.clear();
lcd.print("Booth Available");
}
}

void notifyPriest(String confessionType) {
    digitalWrite(priestLedPin, HIGH);
    tone(priestBuzzerPin, 1300, 800);
    Serial.print("Priest notified: ");
    Serial.println(confessionType);
}

void setBoothStatus(int status) {
    switch (status) {
        case 0:
            digitalWrite(statusGreenPin, HIGH);
            digitalWrite(statusYellowPin, LOW);
            digitalWrite(statusRedPin, LOW);
            break;
        case 1:
            digitalWrite(statusGreenPin, LOW);
            digitalWrite(statusYellowPin, HIGH);
            digitalWrite(statusRedPin, LOW);
            break;
        case 2:
            digitalWrite(statusGreenPin, LOW);
            digitalWrite(statusYellowPin, LOW);
            digitalWrite(statusRedPin, HIGH);
            break;
    }
}

void displayMessage(String message) {
    lcd.clear();
    lcd.print(message);
    delay(2000);
}
```